# Instruction: how to use Bayes_Factors_Context_Effects_Shiny

This Shiny App was built using the experimental data from Dumbalska et al. (2020) (https://www.pnas.org/doi/10.1073/pnas.2005058117.) We re-analyzed the data with QTest, a modeling and statistical software for inequality-constrained statistical inference.

We considered 34 decoy bins that are associated with the attraction effect, the compromise effect, and the similarity effect, based on the Decoy Influence Map provided in the paper. The decoy bins are represented by yellow squares in the attribute map. The Shiny App allows you to choose one or multiple of the 34 bins and specify their effects on the probability of preferring A or B when added as a third option. In addition, it allows you to choose specific probability thresholds for $P_{AB}$ or $P_{BA}$, and test customized context effects predictions. The following describes the steps for testing your hypotheses or the predictions made by a model:

1. Choose a value for the maximum deviation from a deterministic choice. For example, 0.1 implies that the probability of choosing A over B or B over A is between 0.9 and 1, i.e., $0.9 \le P_{AB}\left( or\ P_{BA}\right) \le 1$.
2. Choose a value for the maximum deviation from coin flipping between A and B. For example, 0.1 implies that the probability of choosing A or B is between 0.5-0.1 and 0.5+0.1, i.e.,

   $0.5 - 0.1 \le P_{AB}\left( or\ P_{BA}\right) \le 0.5 + 0.1$.
3. Decide which decoy bins would induce a person to prefer A to B, to be indifferent between A and B, and/or to prefer B to A by selecting the corresponding bins.
4. Click "Run Analysis."

After clicking "Run Analysis," a plot showing the distribution of different Bayes Factor (BF) intervals, along with a qualitative interpretation for each interval, for 188 subjects' choice data appears. The distribution is a visualization of statistical evidence for how many subjects align or don't align with your customized context effects. Underneath the plot is a measure that shows how much your context effects explain all subjects' data jointly. More details on the BF formulations and computations can be found in [ref or link].

You may extract the BF for each individual subject as an .csv file by clicking "Export Bayes Factors." This will be useful if you have different hypotheses or model predictions at hand and want to compete them against each other. As an instance, if you have two specific hypotheses $H_1$ and $H_2$, where

$H_1$ : $0.75 \le P_{AB} \le 1$ (Prefer A) for bin1 and bin2,
$0.75 \le P_{BA} \le 1$ (Prefer B) for bin3 and bin4,
$0.45 \le P_{AB} \le 0.55$ (Indifferent between A and B) for bin5;
$H_2$ : $0.9 \le P_{AB} \le 1$ (Prefer A) for bin33 and bin34,

$0.8 \leq P_{BA} \leq 1$ (Prefer B) for bin31 and bin32,
$P_{AB} = 0.5$ (Indifferent between A and B) for bin30.

After getting the BF of $H_1$ ($BF_{H_1}$) and $H_2$ ($BF_{H_2}$) separately for each individual subject, you can output the BFs to two csv files. In each file, each value is the BF corresponding to subject i's choice data in row i (i = 1 to 188), for the specific hypothesis.

For subject i in row i, the ratio $\frac{BF_{H_1,sub_i}}{BF_{H_2,sub_i}}$ is the BF for $H_1$ against $H_2$ and represents how much more evidence there is for $H_1$ to account for subject i's choice data compared to $H_2$.

Cautions:

1. When choosing the probability thresholds (i.e., values for the maximum deviation from a deterministic choice or coin flipping), make sure that the two intervals do NOT have any overlapping regions (e.g., 0.6<=P<=1 and 0.3<=P<=0.7 have an overlapping interval 0.6<=P<=0.7).
2. When selecting decoy bins, be careful NOT to select the same bin for different predictions (e.g., once "prefer A" for bin1 is chosen, do NOT choose bin1 for "prefer B" or "indifference" between A and B).